

**TỔNG LIÊN ĐOÀN LAO ĐỘNG VIỆT NAM
TRƯỜNG ĐẠI HỌC TÔN ĐỨC THẮNG
KHOA CÔNG NGHỆ THÔNG TIN**



**FINAL REPORT
INFORMATION TECHNOLOGY PROJECT 2**

**BUILDING A WEB APPLICATION TO
SUPPORT EVENT ORGANIZATION
WITH SMART CHECK-IN**

Người hướng dẫn: ThS. Dương Hữu Phước

Người thực hiện: Vũ Quang Huy – 518H0199

Lớp: 18H50203

Khoá: 22

THÀNH PHỐ HỒ CHÍ MINH, NĂM 2025

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ACKNOWLEDGEMENTS

I would like to express my sincere gratitude to Ton Duc Thang University, especially the Faculty of Information Technology, for providing favorable conditions, equipping me with a solid foundation of knowledge, and supporting me throughout my studies and internship

I would like to express my deepest gratitude to Professor Duong Huu Phuoc – my dedicated teacher who has always been ready to listen, guide, and direct me throughout my studies and internship. His dedication and responsibility have helped me better understand my career path and approach problems in a scientific and professional manner.

.

TP. Hồ Chí Minh, ngày 18 tháng 08 năm 2025
Huy
Vũ Quang Huy

ĐỒ ÁN ĐƯỢC HOÀN THÀNH

TẠI TRƯỜNG ĐẠI HỌC TÔN ĐỨC THẮNG

Tôi xin cam đoan đây là sản phẩm đồ án của riêng tôi. Các nội dung nghiên cứu, kết quả trong đề tài này là trung thực và chưa công bố dưới bất kỳ hình thức nào trước đây. Những số liệu trong các bảng biểu phục vụ cho việc phân tích, nhận xét, đánh giá được chính tác giả thu thập từ các nguồn khác nhau có ghi rõ trong phần tài liệu tham khảo.

Ngoài ra, trong đồ án còn sử dụng một số nhận xét, đánh giá cũng như số liệu của các tác giả khác, cơ quan tổ chức khác đều có trích dẫn và chú thích nguồn gốc.

Nếu phát hiện có bất kỳ sự gian lận nào tôi xin hoàn toàn chịu trách nhiệm về nội dung đồ án của mình. Trường đại học Tôn Đức Thắng không liên quan đến những vi phạm tác quyền, bản quyền do tôi gây ra trong quá trình thực hiện (nếu có).

TP. Hồ Chí Minh, ngày 07 tháng 01 năm 2025

Huy

Vũ Quang Huy

PHẦN XÁC NHẬN VÀ ĐÁNH GIÁ CỦA GIẢNG VIÊN

Phần xác nhận của GV hướng dẫn

Tp. Hồ Chí Minh, ngày tháng năm
(kí và ghi họ tên)

Phần đánh giá của GV chấm bài

Tp. Hồ Chí Minh, ngày tháng năm
(kí và ghi họ tên)

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CHAPTER 1. PROJECT INTRODUCTION

1.1 Project Rationale

The project "Building a Web Application to Support Event Organization with Smart Check-in" addresses the practical needs of the current market. Organizations and schools often face difficulties in managing events and conducting manual attendance checks, leading to wasted time and errors. This application not only provides attendance tracking using QR codes but also integrates an automated attendance system, saving users time and minimizing errors. Simultaneously, this project offers an excellent opportunity to apply popular technologies in full-stack web development (React, Spring Boot, QR code scanning) and learn about software architecture, security, and UX/UI design. With its moderate complexity and high practicality, this project is a suitable choice for learning and developing professional programming skills

1.2 Project Objectives

The objective of the project " Building a Web Application to Support Event Organization with Smart Check-in " is to create a comprehensive management system that handles event information, registration, and event change tracking. The system will help store and manage event information efficiently, minimizing errors during organization, while automating processes such as registration, guest list management, and tracking attendance times and actions. Furthermore, the system aims to save time and resources, improving management efficiency for individuals and organizations. The product not only meets practical needs but also provides a more professional, modern, and convenient event management experience.

1.3 Scope of the Study

The project focuses on building the core functionalities of an event management system, including: event management, a smart check-in system using QR codes, user management, as well as additional features such as

event registration, comments, resource uploads, and invitation management. The application is designed with a responsive interface across different devices, ensuring security through authentication, authorization, and data encryption, along with an optimally designed and thoroughly tested relational database.

1.4 Research Methodology

To gain a solid understanding of the technologies used in the project, the team conducted a user needs analysis to identify requirements for the application and investigated whether React and Spring technologies could adequately support the project. Based on the research results, appropriate technologies were selected and applied in the system development process. The backend was implemented using Spring Boot with frameworks and libraries such as Spring Security (authentication), Spring Data JPA (database ORM), Spring Web (REST API development), Lombok (boilerplate code reduction), and OAuth2 integration. The frontend was developed using React, along with supporting libraries including Axios (API calls), React Router (navigation), and QR-code/jsQR (QR code scanning).

1.5 Practical Significance

To optimize the event organization process, reduce manual workloads, enhance user experience, and apply modern technologies to event management in a professional and efficient manner.

CHAPTER 2. REQUIREMENTS ANALYSIS

2.1 Basic Requirements

The Event Management system provides essential tools for organizers and participants.

As an organizer, users can create events, choosing the type of event (public or school-specific). They can provide complete information such as event name, date, start time, duration, event category, event mode (offline, online, or hybrid), number of registrants, Q&A link, and event participation link (if online or hybrid), location, event description, and event photos. They can also choose attendance methods (no attendance check, automatic system attendance, or QR code scanning) and add speaker information.

After an event is created, if the organizer wants people to register, it must be reviewed by the admin. The organizer can request event approval to open registration.

Additionally, the organizer can manage event resources such as editing the event, editing speakers, uploading materials, and interacting with comments on the event. The organizers can track the attendance history of the event. They can invite people to participate in the event through invitations available on the system.

The core function of the system is that the organizers can conduct attendance by scanning QR codes or setting a time for the system to take attendance.

As event participants, users can register for the event to create tickets. They receive notifications about event changes via email. In addition, registered participants can view comments and download necessary documents. Event participants can check in automatically when the time comes, with the system integrating location verification to prevent fraud, or participants can check in using a QR code sent via email.

Administrators can approve events, manage user accounts, and create appropriate categories for the event.

2.2 Actor

STT	Actor	Description
1	Admin	- User Management (accounts, registration history, check-in/attendance history, event creation history), - Category Management (school categories, event categories), - Event Management (event approval, event cancellation or deletion).
2	Teacher	Create, update, and delete events; submit events for approval; invite participants; perform attendance check-in using QR code scanning; upload related documents; respond to participant comments; and reopen the check-in period when necessary.
3	Student	Register for events; receive email notifications for registered events; track event schedules; confirm participation; perform attendance check-in via QR code scanning or automatic check-in; download related materials; and post comments on events.

2.3 Use case description

ID	Usecase	Description
UC01	Sign in	Users log into the system using their Google account.
UC02	Sign out	Users should log out when not using the system to ensure security.
UC03	View profile	Users can view and edit their personal information.
UC04	View the list of upcoming events.	Users can view a list of upcoming events.
UC05	View the event invitation.	Users can view event invitations.
UC06	View the list of events for today's attendance check.	Users can view today's events and attendance times on the system.
UC07	View event details	Participants can view event details, including the schedule, speakers, and attendance times, on the system if available.
UC08	Register to participate in the event.	Users register to participate in the event.
UC09	View attendance history	Users can review their attendance history.
UC010	Management resource	Event document management allows users to download and upload documents.
UC011	Comment on event	Helps users exchange event information more easily.
UC012	Check-in to the event	Participants can check in via QR code or through the system, depending on the event's configuration.
UC013	Check out of the event	Participants can check out via QR code or through the system, depending on the event's configuration.
UC014	Responding to event invitations	Participants accept or decline the invitation from the organizers.

UC015	Create new event	Create a new event and set up the information, QR code access mode, or AUTO mode for the event.
UC016	Edit event	The organizers will update event information, add smaller events, or change the schedule.
UC017	Confirm the event	Admins review events before publishing them.
UC018	Cancel the event	The organizers want to cancel the event
UC019	Open register events	The organizers want to open the event for all everyone register
UC020	User account management	Admins can lock, delete, and change the roles of accounts.
UC021	View history register event	Admins can view event registration history.
UC022	View history check-in/out	Admin can view history check-in/out
UC023	View history create event	Admin can view history create event
UC024	School catalog management	Admin can create school catalog
UC025	Event catalog management	Admin can create event catalog
UC026	Export event attendance list	The teacher can export the check-in/out file.

CHAPTER 3. SYSTEM DESIGN

3.1 System architecture

3.1.1 Client-Server Modal

The event management system is built on a Client-Server model with a separate Frontend and Backend architecture. Specific:

Frontend (ReactJs): Develops the user interface (UI/UX) and handles user-system interaction.

Backend (Node Js): Provides APIs, communicates with the database, authenticates users, manages event information, and performs attendance tracking.

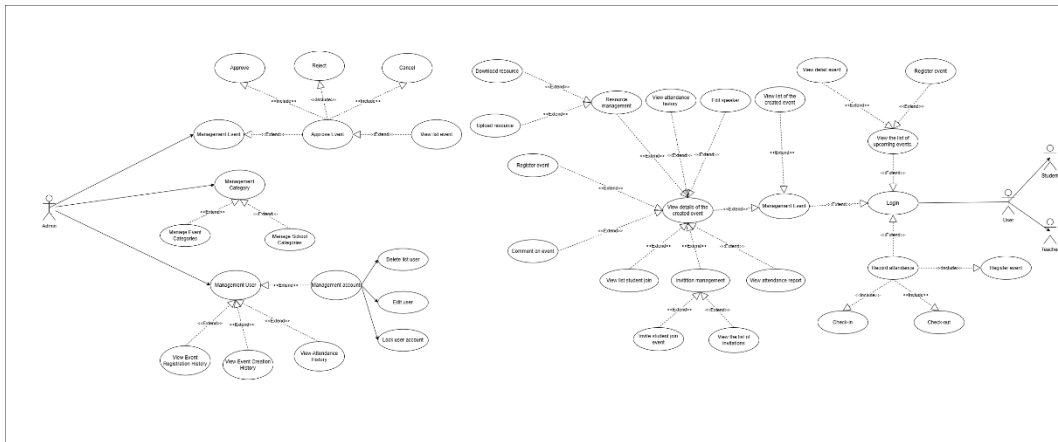
Database (MySQL): Stores user information, events, tickets, and records attendance history.

API connection: Communication between frontend and backend via REST API.

3.1.2 Reason for choosing the client-server modal

The client-server model was chosen for the project because it clearly separates the user interface (React) from the business logic processing (Spring Boot), thus providing high security as authentication and authorization are performed on the server. This model allows for flexible scalability to support thousands of concurrent users, optimizes performance through caching, and centralizes data processing on the server. It also offers a high degree of flexibility as a single backend server can serve multiple different clients (web, mobile), is easy to maintain as the server can be upgraded without affecting clients, and adheres to REST API standards for standardized communication. This model is particularly well-suited to the requirements of an intelligent login system as it demands robust authentication, strict access control, and server-side attendance statistics processing.

3.2 Use case diagram



3.3 Đặc tả usecase

3.3.1 Usecase Sign in

Bảng 3.1 Usecase Sign in

Use case ID	UC01	
Use case Name	Sign in	
Description	The user wants to login the system to use the function of system	
Actor(s)	Admin, Teacher, Student	
Trigger	The user want to login the system	
Pre-conditions	Users must have an account registered on the system.	
Post-conditions	Users can log into the system and use the function on the system	
Basic flow	Actor	System
	1. The user logs into the system using their Google account. 2. Select the Google account.	1. The system displays the login interface. 2.1 The system checks account information. 2.2 The system's main screen is displayed.

3.3.2 Usecase Sign out

Bảng 3.2 Usecase Sign out

Use case ID	UC02	
Use case Name	Sign out	
Description	The user want to sign out the system	
Actor(s)	Admin, Teacher, Student	
Trigger	The user want to sign out the system	
Pre-conditions	The user has successfully logged into the system.	
Post-conditions	User successfully sign out	
Basic flow	Actor	System
	1. The user click “Sign out” function	1. The system logs you out of your account and returns you to the login screen.

3.3.3 Usecase View profile

Bảng 3.3 Usecase View profile

Use case ID	UC03	
Use case Name	View profile	
Description	When users need to check or update their personal information.	
Actor(s)	Admin, Teacher, Student	
Trigger	The user want to change profile	
Pre-conditions	The user has successfully logged into the system.	
Post-conditions	Display personal information, allowing editing (if needed).	
Basic flow	Actor	System
	1. The user click to “Profile” 2.Editing information (if needed)	1. Display UI profile 2. Save information

3.3.4 Usecase View the list of upcoming events.

Bảng 3.4 Usecase View the list of upcoming events.

Use case ID	UC03	
Use case Name	View the list of upcoming events.	
Description	Users want to see upcoming events.	
Actor(s)	Student, Teacher	
Trigger	The user want to view the list of upcoming events	
Pre-conditions	The user has successfully logged into the system. The user had previously registered for the event.	
Post-conditions	Display a list of upcoming events.	
Basic flow	Actor	System
	1. Login the system. 2. In the home page click function upcoming events	2.1 Filter upcoming events. 2.2 Display list upcoming events

3.3.5 Usecase View the event invitation

Bảng 3.5 Usecase View the event invitation.

Use case ID	UC05	
Use case Name	View the event invitation..	
Description	Users want to view event invitations.	
Actor(s)	Student, Teacher	
Trigger	The user want to view event invitation	
Pre-conditions	The user has successfully logged into the system.	
Post-conditions	Display the list event invitation	
Basic flow	Actor	System
	1. Login the system. 2. In the home page click function invitation	2.1 Filter upcoming events. 2.2 Display list event invitation (if have)

3.3.6 Usecase View the list of events for today's attendance check.

Bảng 3.6 Usecase View the list of events for today's attendance check.

Use case ID	UC06
Use case Name	View the list of events for today's attendance check.

Description	The user want to view or attendance check for the event.	
Actor(s)	Student, Teacher	
Trigger	The user want to view the list of events for today's attendance check	
Pre-conditions	The user has successfully logged into the system.	
Post-conditions	Display personal information, allowing editing (if needed).	
Basic flow	Actor	System
	1. Login the system. 2. In the home page click function today's attendance	2.1 Filter today's attendance event 2.2 Display list event today's attendance

3.3.7 Usecase View event details

Bảng 3.7 Usecase View event details

Use case ID	UC07	
Use case Name	View event details.	
Description	Users want to see detailed information such as the time, location, etc.	
Actor(s)	Teacher, Student	
Trigger	The user want to view event details	
Pre-conditions	The user has successfully logged into the system.	
Post-conditions	Display information detail of the evetn	
Basic flow	Actor	System
	1. Login the system. 2. Click events. 3. Click button view detail	2 Display all upcoming events. 3 Display event detail

3.3.8 Usecase Register to participate in the event.

Bảng 3.8 Usecase Register to participate in the event.

Use case ID	UC08	
Use case Name	Register to participate in the event.	
Description	Users register to participate in the event..	
Actor(s)	Teacher, Student	
Trigger	Users want to participate in the event.	
Pre-conditions	Users must successfully log into the system. The event has not reached its registration limit.	
Post-conditions	User register event success Receive the email register success	
Basic flow	Actor	System
	1. Click button event detail. 2. Click register button	1. Display UI view detail event 2. The system acknowledges successful registration and sends a registration confirmation email.

3.3.9 Usecase View attendance history

Bảng 3.9 Usecase View attendance history

Use case ID	UC09	
Use case Name	View attendance history	
Description	When user want to check record the attendance time	
Actor(s)	Teacher, Student	
Trigger	The user want to view attendance history	
Pre-conditions	User must register event	
Post-conditions	Display the time attendance of user	
Basic flow	Actor	System
	1. The user click button history	1.Display attendance time record

3.3.10 Usecase Management resource

Bảng 3.10 Usecase Management resource

Use case ID	UC010	
Use case Name	Management resource	
Description	Event document management allows users to download and upload documents.	
Actor(s)	Teacher, Student	
Trigger	When user wan to upload, download or delete the document	
Pre-conditions	User create the event to (upload) or register the event to download (student)	
Post-conditions	User upload or download success	
Basic flow	Actor	System
	1. The tab resource on the event. 2. Teacher: click button choose file and click upload Student: download the resource on the table	1. Display UI resource 2. Upload file success display file update on the table Download to the computer of the user

3.3.11 Usecase Comment on event

Bảng 3.11 Usecase Comment on event

Use case ID	UC011	
Use case Name	Comment on event	
Description	Helps users exchange event information more easily.	
Actor(s)	Teacher, Student	
Trigger	Users want to discuss the event.	
Pre-conditions	Teacher: Create the event Student: Register the event	
Post-conditions	User can comment on the event	
Basic flow	Actor	System
	1. The tab comment on the event. 2. Teacher: click button choose file and click upload	1.1 Load all comment of the event 1.2 Display UI with comment

	Student: download the resource on the table	
--	---	--

3.3.12 Usecase *Check-in to the event*

Bảng 3.12 Usecase Check-in to the event

Use case ID	UC012	
Use case Name	Check-in to the event	
Description	Participants can check in via QR code or through the system, depending on the event's configuration.	
Actor(s)	Student, Teacher	
Trigger	User want to check-in the event	
Pre-conditions	The user register the event successfully	
Post-conditions	User can check-in when on time of the event	
Basic flow	Actor	System
	Auto mode: 1. Click the tab today attendance 2. Click button check-in 3. Click button check-out Qr code: 1. Show the Qr code on the camera	Auto mode: 1. Display UI attendance 2.1 Check location of user is valid 2.2 Record the check-in time and unlock the button check-out when on time. Display record the check-in time Qr code: 1. The system will record the time if the Qr code is correct. 1.1 Show the information of the user on the screen

3.3.13 Usecase *Check out of the event*

Bảng 3.13 Usecase Check out of the event

Use case ID	UC013
Use case Name	Check out of the event
Description	Participants can check out via QR code or through the system, depending on the event's configuration..

Actor(s)	Student, Teacher	
Trigger	The user want to check out of the event	
Pre-conditions	The user must check-in the event first	
Post-conditions	Display button check-out	
Basic flow	Actor	System
	1. Click button check-out	1. Record check-out time 2. Show the check-out time on UI

3.3.14 Usecase Responding to event invitations

Bảng 3.14 Usecase Responding to event invitations

Use case ID	UC014	
Use case Name	Responding to event invitations	
Description	Participants accept or decline the invitation from the organizers	
Actor(s)	Student, Teacher	
Trigger	The user want to response the event invitation	
Pre-conditions	The user already have account on the system The teacher will send the invitation	
Post-conditions	User can accept or decline the invitaion	
Basic flow	Actor	System
	1. Click function invitation 2.1 Click button accept 2.2 Click button reject	1. Display UI invitation 2.1.1 The user will receive the email accept the event. 2.2.2 Update status and record the status user.

3.3.15 Usecase Create new event

Bảng 3.15 Usecase Create new event

Use case ID	UC015	
Use case Name	Create new event	
Description	Create a new event and set up the information, QR code access mode, or AUTO mode for the event.	
Actor(s)	Admin, Teacher	
Trigger	The user wanto create new event	
Pre-conditions	Users must log into the system.	

	User accounts must have appropriate permissions.	
Post-conditions	Users can access the event creation page.	
Basic flow	Actor	System
	1. Select Create New Event 2. The user chooses the event type: public or for school students. Click Next. 3. The user provides all required information (e.g., event name, time, location, check-in/check-out method). Click the Create button. 4. The user can skip this step or create a speaker.	1.1. Event creation page display system 2.1. Event creation form display system 3.1. The system starts creating a new event from the provided information. 3.2. Displays a successful creation message. 3.3 Displays the modal for creating speakers for the event.

3.3.16 Usecase Edit event

Bảng 3.16 Usecase Edit event

Use case ID	UC016	
Use case Name	Edit event	
Description	The organizers will update event information, add smaller events, or change the schedule..	
Actor(s)	Teacher, Admin	
Trigger	The user want to change information of the event	
Pre-conditions	The event created	
Post-conditions	Display the information of the event on edit mode	
Basic flow	Actor	System
	1. Click detail event 2. Click button edit 3. Fill information edit 4. Click button save	2. Load data and display modal edit 3. Save the information changed

3.3.17 Usecase Confirm the event

Bảng 3.17 Usecase Confirm the event

Use case ID	UC017	
Use case Name	Confirm the event	
Description	The administrator wants to test the event before making it open for public registration..	
Actor(s)	Admin	
Trigger	The administrator wants to moderate the event.	
Pre-conditions	Events must be submitted for review.	
Post-conditions	Administrators can review or reject registrations for public events.	
Basic flow	Actor	System
	1. Click the approve/reject button.	1.1 The system will update the event status. 1.2 The system will enable public registration if the admin approves the event.

3.3.18 Usecase Cancel the event

Bảng 3.18 Usecase Cancel the event

Use case ID	UC018	
Use case Name	Cancel the event	
Description	The user wants to cancel the event due to some kind of problem.	
Actor(s)	Teacher, Admin	
Trigger	The user wants to cancel the event	
Pre-conditions	Canceled events cannot be ongoing events.	
Post-conditions	Users can cancel an event and notify participants.	
Basic flow	Actor	System
	1. The user accesses the event management page. 2. Clicks the cancel event button.	2.1Event cancellation successful. 2.2Email notification sent to participants.

3.3.19 Usecase Open register events

Bảng 3.19 Usecase Open register events

Use case ID	UC019	
Use case Name	Open register events	
Description	Users want to invite others to an event without making it public.	
Actor(s)	Teacher	
Trigger	The user wants to open register events	
Pre-conditions	User accounts must be registered. Events must be approved before invitations can be sent. Events must be in progress.	
Post-conditions	Users can send invitations to members within the system.	
Basic flow	Actor	System
	1. Select an approved event. 2. Click on the invitation tab, enter the email address to invite. 3. Click the send button.	1.1 The system will display the invitation interface. 2.1 The system will search for emails. 3.1 The system will send invitations via email.

3.3.20 Usecase User account management

Bảng 3.20 Usecase User account management

Use case ID	UC020	
Use case Name	User account management	
Description	Admins can lock, delete, and change the roles of accounts.	
Actor(s)	Admin	
Trigger	Admin want change the account of users	
Pre-conditions	User already register in the system	
Post-conditions	Admin can lock, delet or change the account	
Basic flow	Actor	System
	1. Click function account management	1.Display UI account managemnet 2. Save data

	2. Admin can lock, delete or change the account of user.	
--	--	--

3.3.21 Usecase View history register event

Bảng 3.21 Usecase View history register event

Use case ID	UC021	
Use case Name	View history register event	
Description	Admins can view all event registration history.	
Actor(s)	Admin	
Trigger	The user want to change profile	
Pre-conditions	The user has successfully logged into the system.	
Post-conditions	Display personal information, allowing editing (if needed).	
Basic flow	Actor	System
	1. The user click to “Profile” 2.Editing information (if needed)	1. Display UI profile 2. Save information

3.4 Activity Diagram

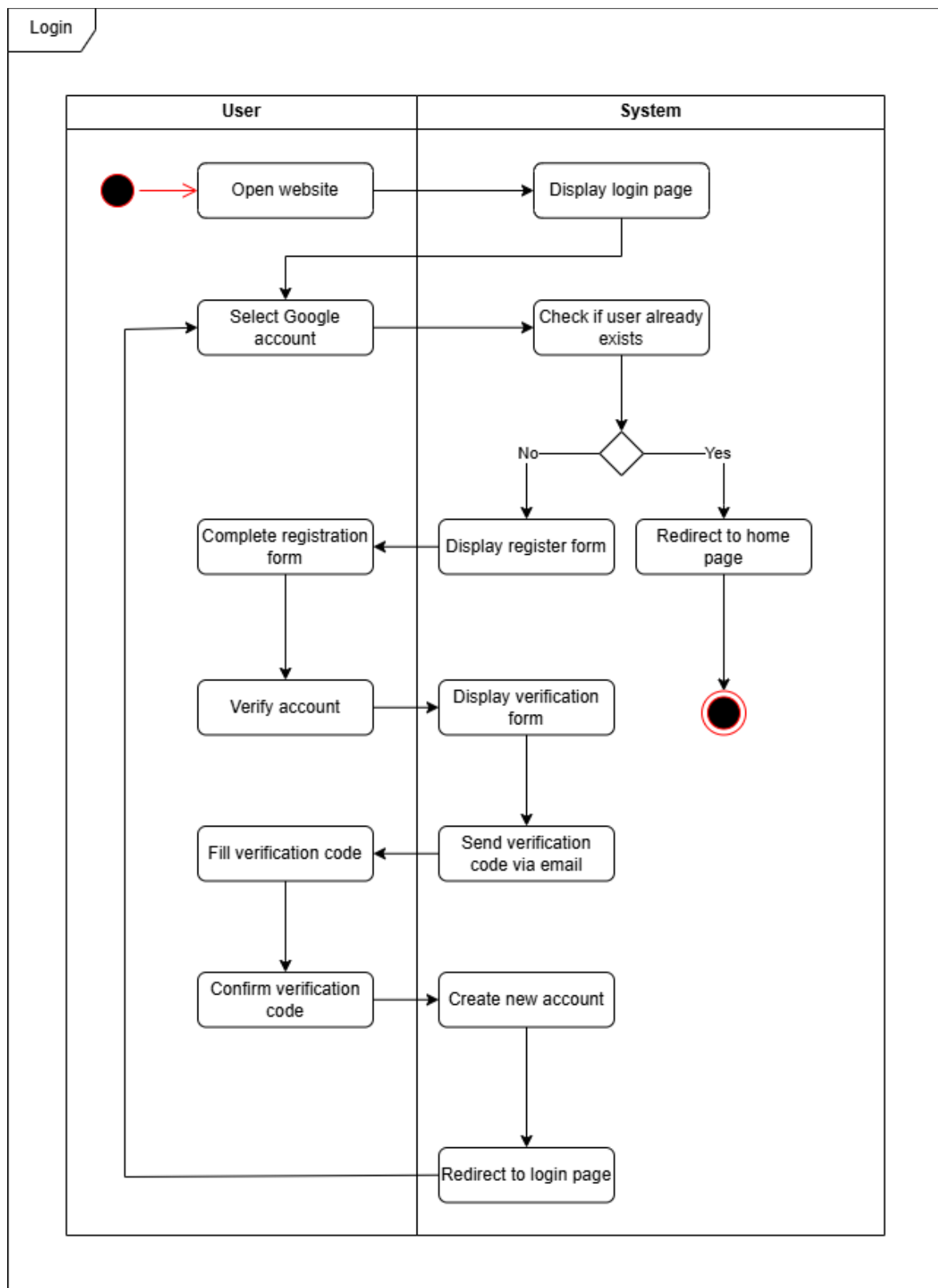


Figure 4.1. Activity diagram login

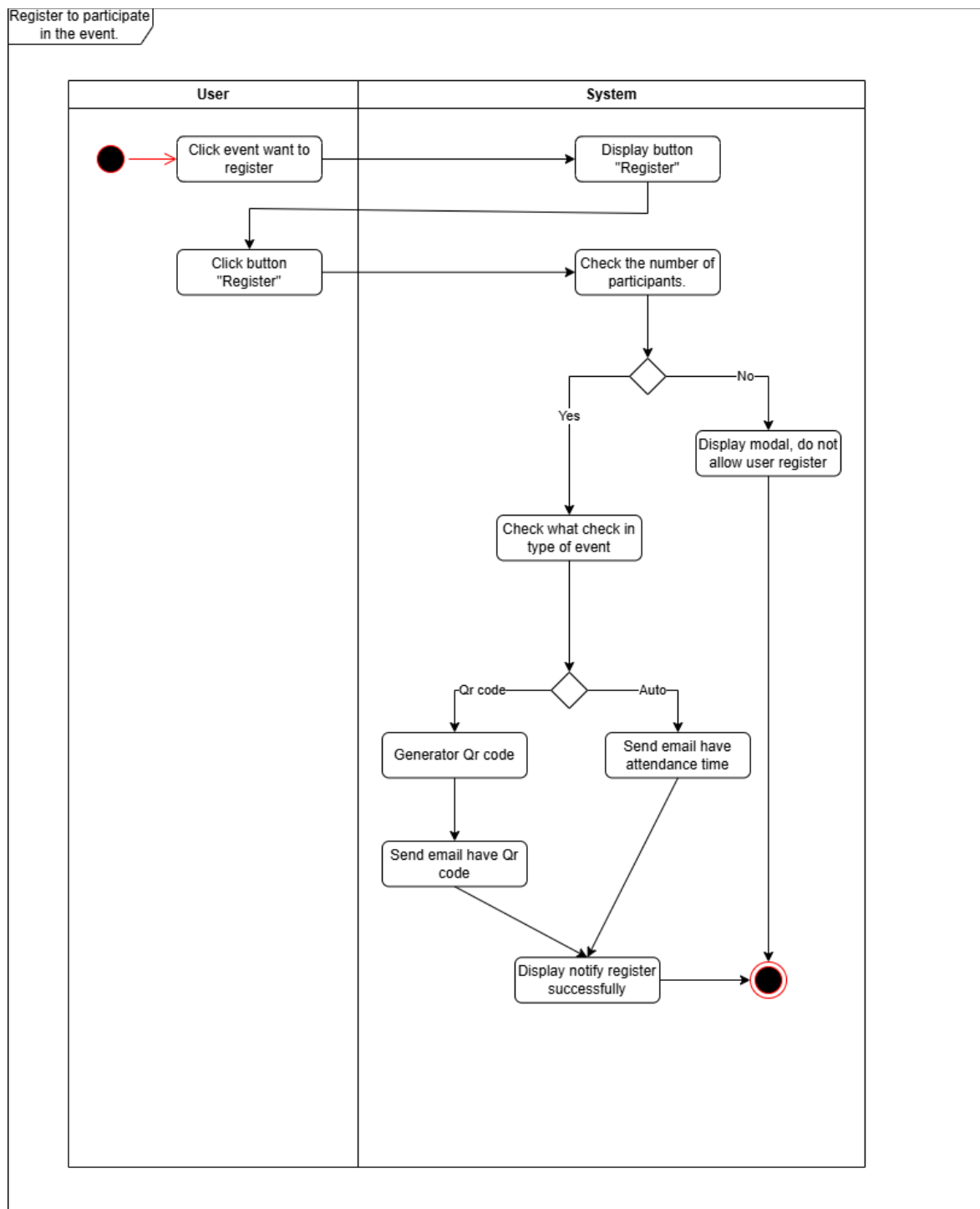


Figure 4.2. Activity diagram register the event

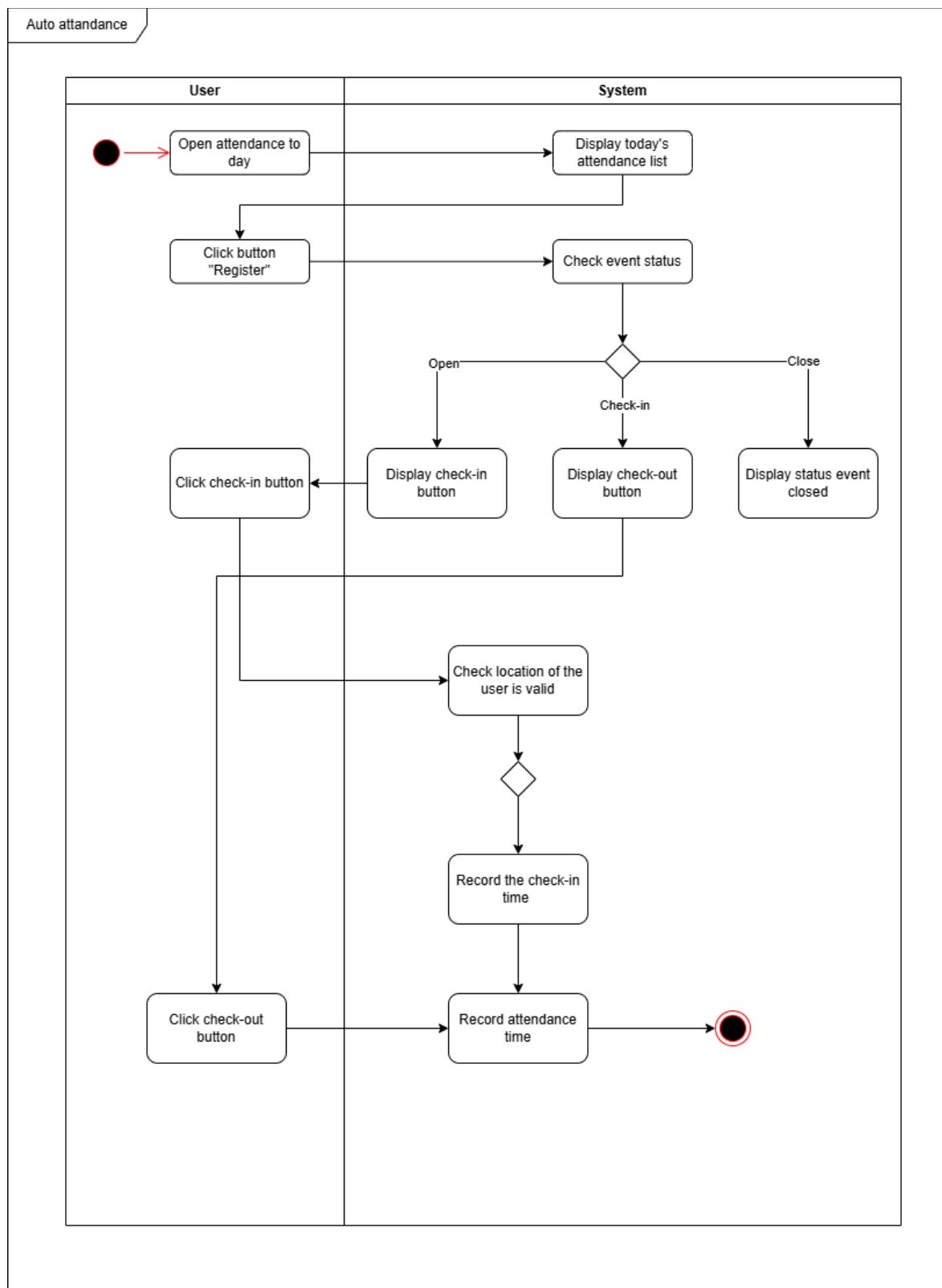


Figure 4.3. Activity diagram auto attendance

3.5 Entity relationship diagram

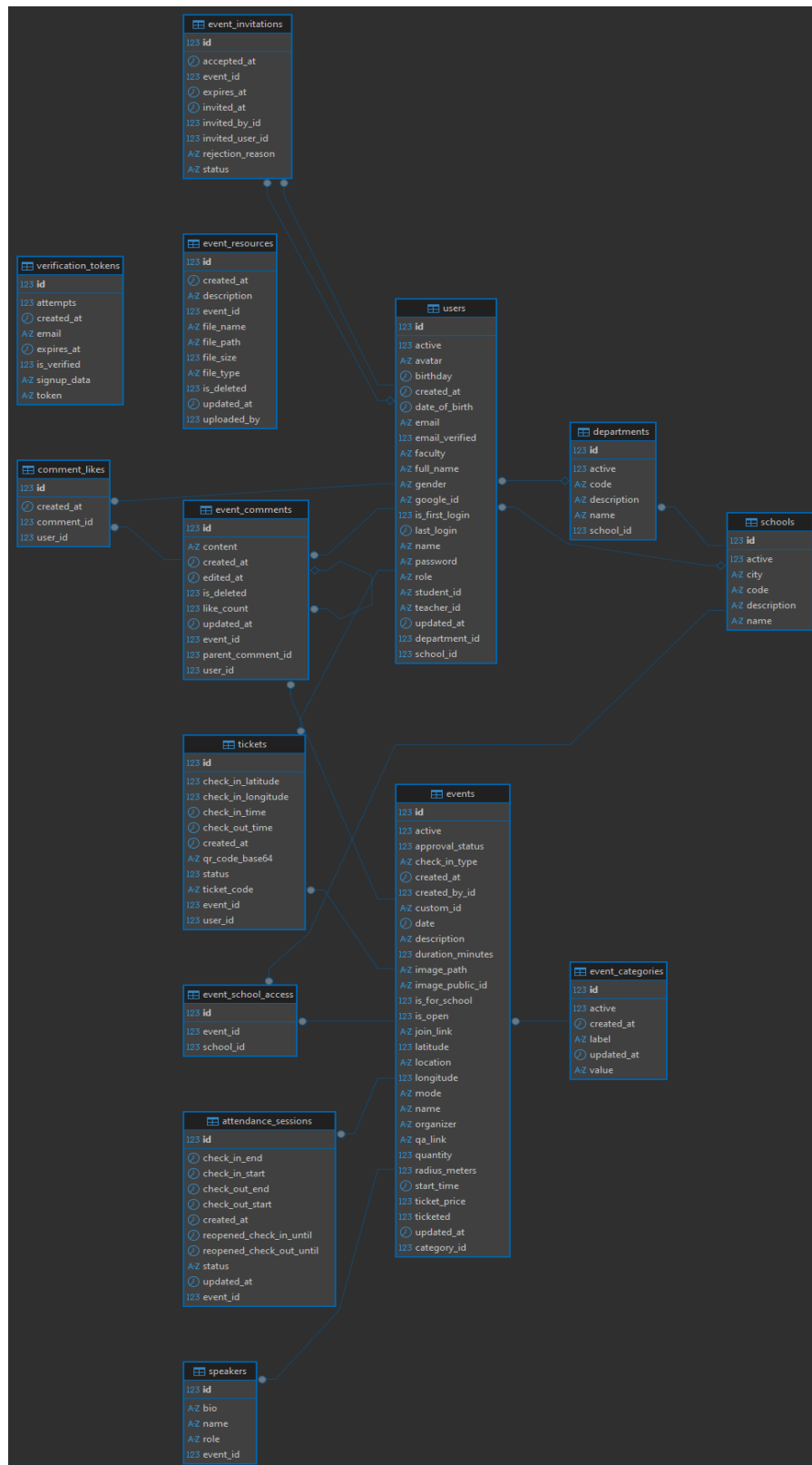


Figure 5.1. Entity relation diagram

CHAPTER 4. KHAI ỨNG DỤNG

4.1 Login page

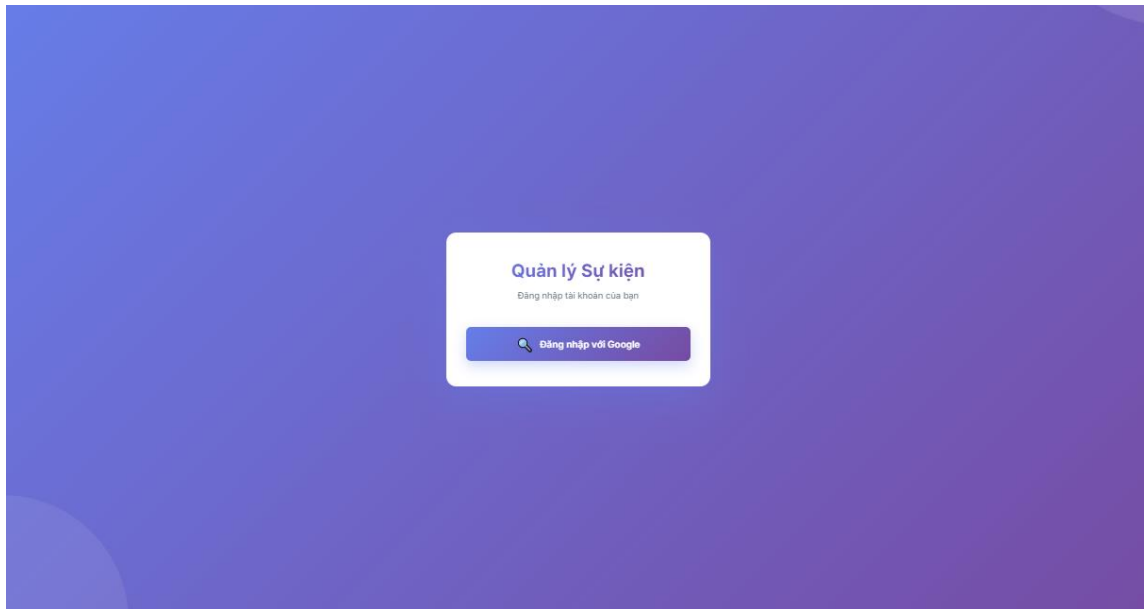


Figure 4.1: Login page

Users access the system via Google Login. Users use the login button to select their Google account to log in. If the user does not have an account on the system, they will be redirected to the Google account verification page to complete the login process.

Users can track notifications for registered events, such as upcoming event times, event invitations, or events scheduled for today.

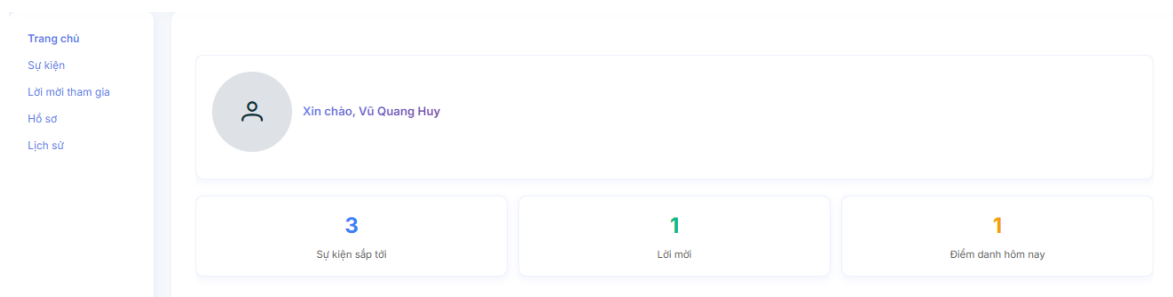


Figure 4.2 Home page

4.2 User account management

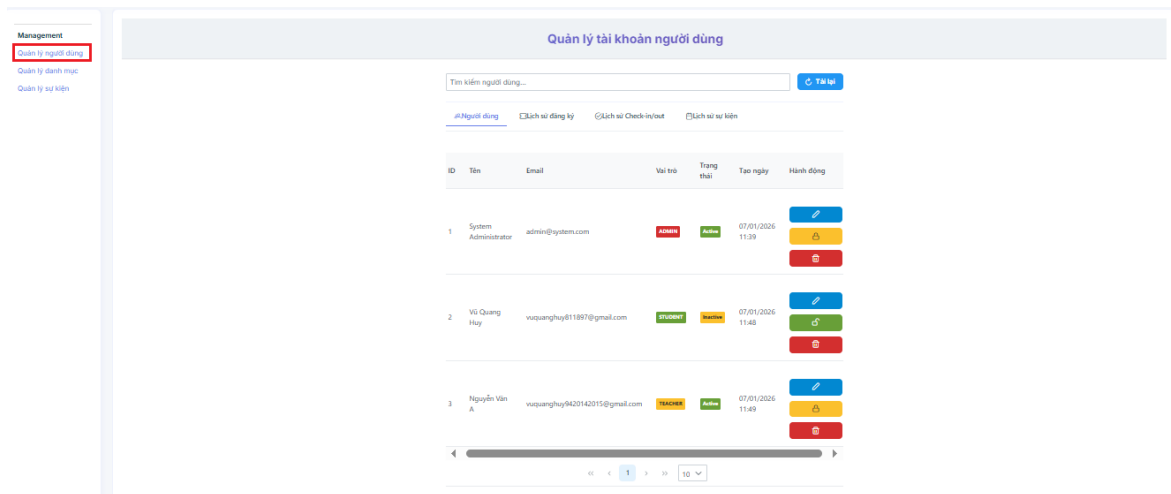


Figure 4.3 User account management

Admins can access the user management section to view the user list, edit information, and update account status.

4.3 Create new event

The "Add New Event" function allows users (usually teachers or administrators) to create a new event on the system. Event type, event name, date, start time, category, number of participants, etc., can be selected. Additionally, the form supports attendance tracking methods; users can choose between automatic attendance tracking or QR code scanning.

Tạo sự kiện mới

Loại sự kiện:  Công khai

Tên sự kiện

Ngày tổ chức

09-01-2026



Thời gian bắt đầu (HH:mm)

13:43



Thời lượng (phút)

60

Danh mục

Hội thảo

Chế độ

Offline



Số lượng người đăng ký

Để trống = không giới hạn

Link hỏi đáp

Địa điểm

Nhập địa chỉ sự kiện

Mô tả

Ảnh sự kiện (tùy chọn)

Choose File

No file chosen

Lựa chọn phương thức điểm danh

Không điểm danh



Tạo

Hủy

Figure 4.4 Create new event

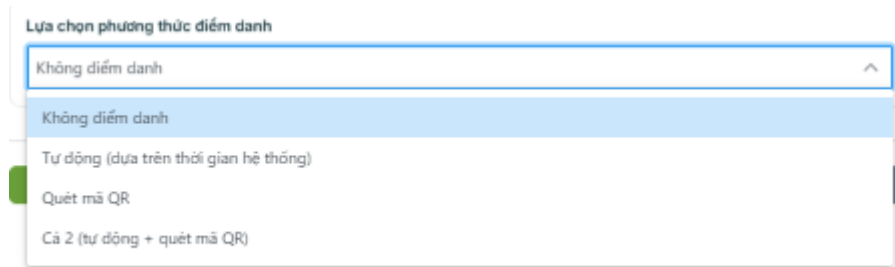


Figure 4.5 Select method attendance

4.4 Page event management

Quản lý sự kiện							
Tải lại							
Mã	Tên Sự Kiện ↑↓	Ngày diễn ra ↑↓	Hình Thức ↑↓	Trạng Thái ↑↓	Trạng Thái Duyệt ↑↓	Mở Đăng Ký	Hành Động
1	Leadership Skills Workshop	09/01/2026	OFFLINE	Sắp diễn ra	Chưa gửi	☐ Đóng	...
2	Startup Ideas Workshop	09/01/2026	OFFLINE	Sắp diễn ra	Gửi duyệt	☐ Đóng	...
3	Workshop Lập trình Web	13/01/2026	HYBRID	Sắp diễn ra	Từ chối	☐ Đóng	...
4	Full-Stack Development Workshop	09/01/2026	ONLINE	Sắp diễn ra	Chấp thuận	☒ Mở	...
<< < 1 > >> 10 ▾							

Figure 4.6 Event management

The event management page allows event administrators or creators to view detailed event information.

Quản lý sự kiện							
Tải lại							
Mã	Tên Sự Kiện ↑↓	Ngày diễn ra ↑↓	Hình Thức ↑↓	Trạng Thái ↑↓	Trạng Thái Duyệt ↑↓	Mở Đăng Ký	Hành Động
1	Leadership Skills Workshop	09/01/2026	OFFLINE	Sắp diễn ra	Chấp thuận	☒ Mở	⋮
2	Startup Ideas Workshop	09/01/2026	OFFLINE	Sắp diễn ra	Chấp thuận		
3	Workshop Lập trình Web	13/01/2026	HYBRID	Sắp diễn ra	Chấp thuận		
4	Full-Stack Development Workshop	07/01/2026	ONLINE	Đang diễn ra	Chấp thuận		
<< < 1 > >> 10				Mở lại điểm danh Quét mã điểm danh Gửi duyệt Hủy sự kiện Xóa sự kiện			

Figure 4.7 Function support for teacher

Additionally, the event management page has user-friendly functions. The admin event page includes options to accept, reject, cancel, or delete events.

Quản lý sự kiện							
Tìm kiếm							
MG	Tên Sự Kiện [v]	Ngày diễn ra [v]	Hình Thức [v]	Trạng Thái [v]	Trạng Thái Duyệt [v]	Mô Đăng Ký	Hành Động
1	Leadership Skills Workshop	06/01/2026	OFFLINE	Sắp diễn ra	Chấp thuận	☒ MÔ	...
2	Startup Ideas Workshop	06/01/2026	OFFLINE	Sắp diễn ra	Chấp thuận	☐ MÔ	✓ Chấp nhận ✗ Từ chối
3	Workshop Lập trình Web	13/01/2026	HYBRID	Sắp diễn ra	Chấp thuận	☒ MÔ	⚙️ Hủy sự kiện 🗑️ Xóa sự kiện
4	Full-Stack Development Workshop	07/01/2026	ONLINE	Sắp diễn ra	Chấp thuận	☒ MÔ	

Figure 4.8. Function support for admin

4.5 Page view detail event

The Event Information tab allows users to view detailed event information. This includes the event name, location, attendance method, etc.

The Edit Event button is displayed for the event owner or administrator.

The Speaker tab allows you to view speaker information, including their role, name, and description.

The speaker management button is only visible to event administrators and owners. It allows users to add, delete, or edit speaker information.

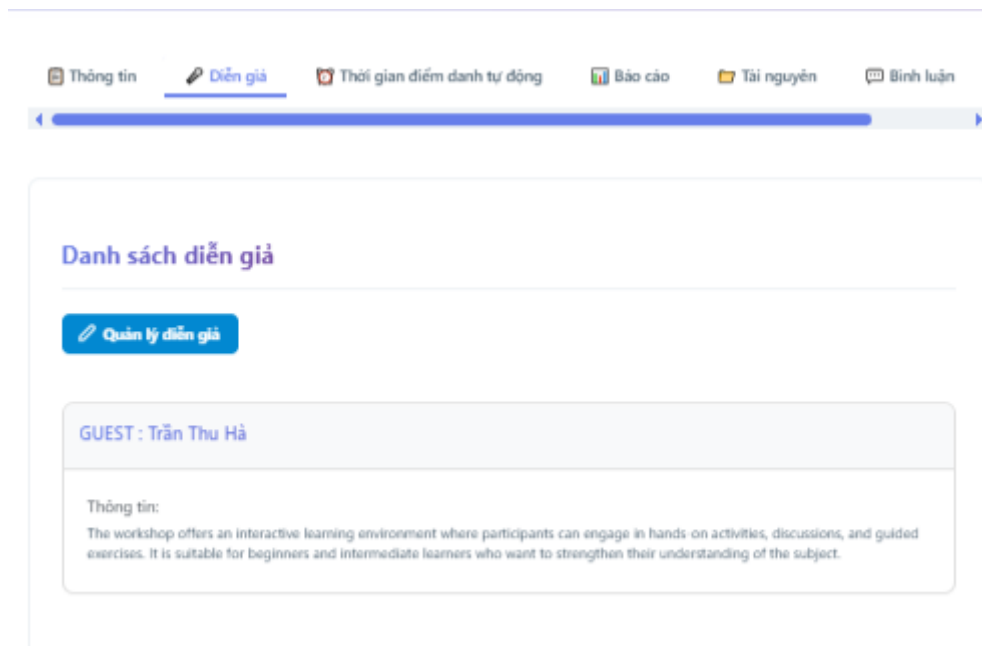
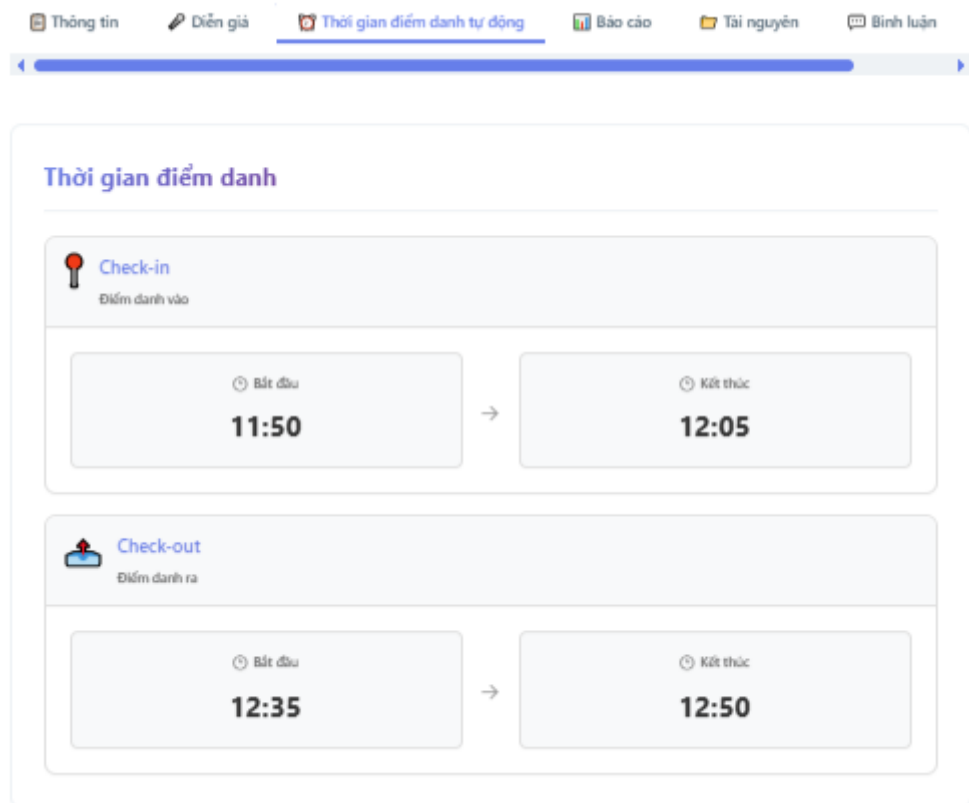
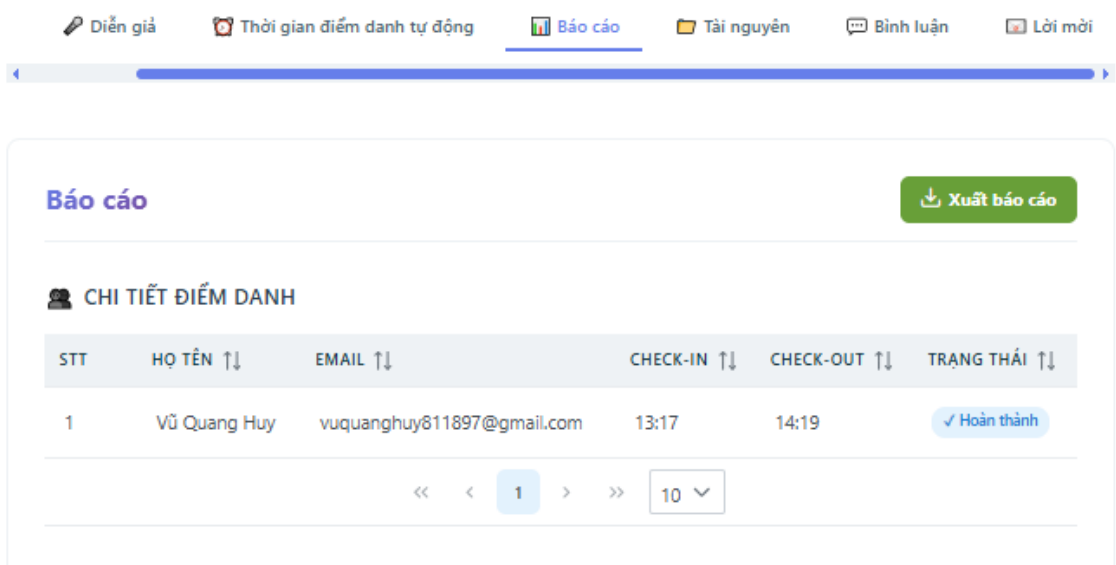


Figure 4.10 View Detail speaker

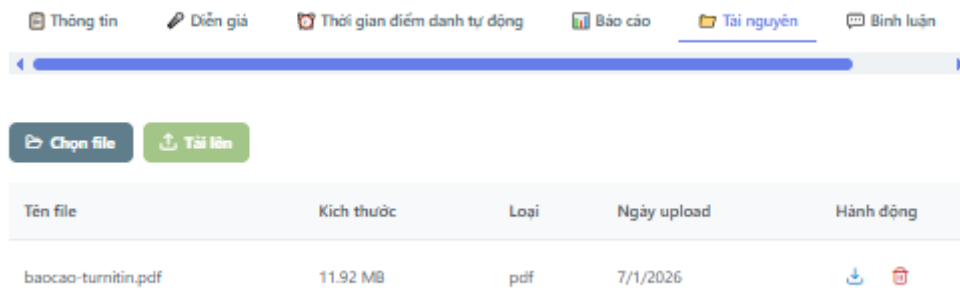
The automatic attendance time tab allows you to set the check-in and check-out times. It only displays for events configured as auto.



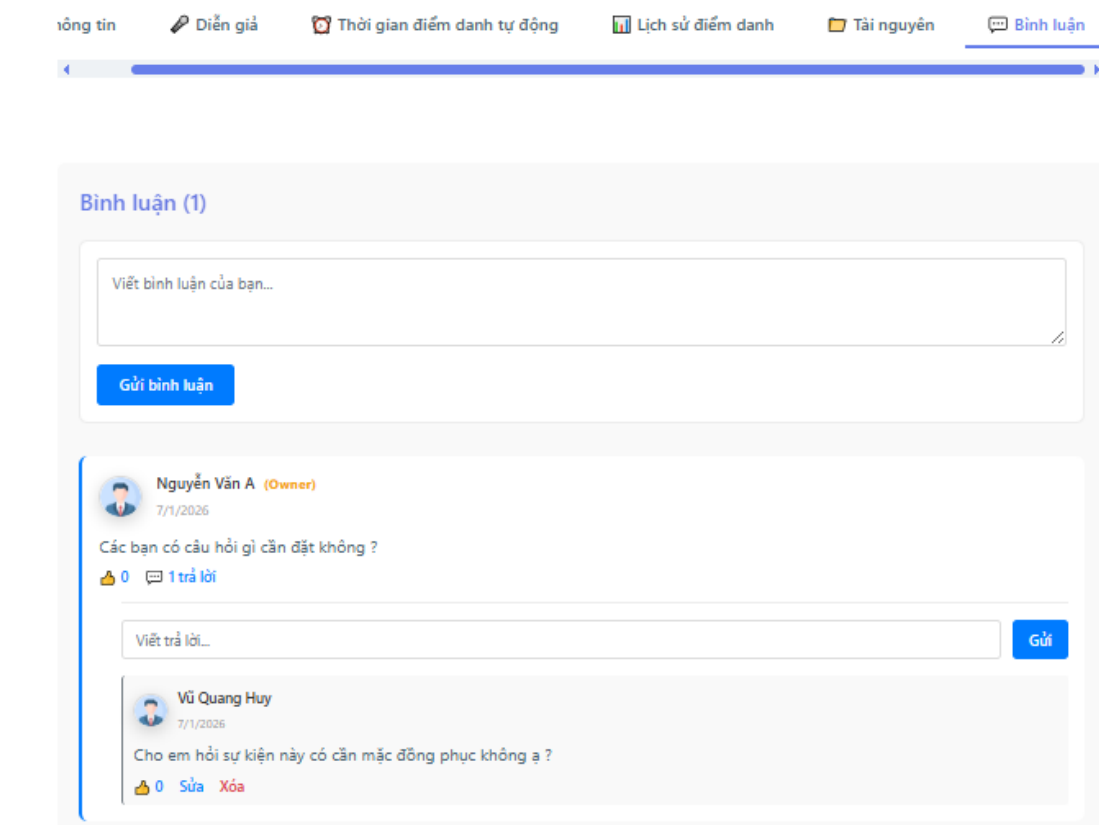
The report allows users to easily view their check-in and check-out history. It displays essential information such as name, time, and status.



Allows users to upload course materials to the system. Teachers can upload materials and registered users can download them.



The commenting function allows people to interact with each other and resolve any remaining questions.



Event organizers can invite people to attend an event if they don't want to publicize it.

Điền giá

Thời gian điểm danh tự động

Báo cáo

Tài nguyên

Bình luận

Lời mời

Gửi lời mời tham gia (Email)

Nhập email người muốn mời

Gửi mời

Danh sách lời mời (1)

Email	Ngày mời	Trạng thái	Lý do từ chối	Hành động
vuquanghuy811897@gmail.com	07/01/2026 12:40	Chờ xác nhận	-	

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Invited guests may accept or decline the invitation. Please state your reason for being unable to attend the event, if applicable.

Lời Mời Sự Kiện

LỜI MỜI (1)

Startup Ideas Workshop

Chờ xử lý

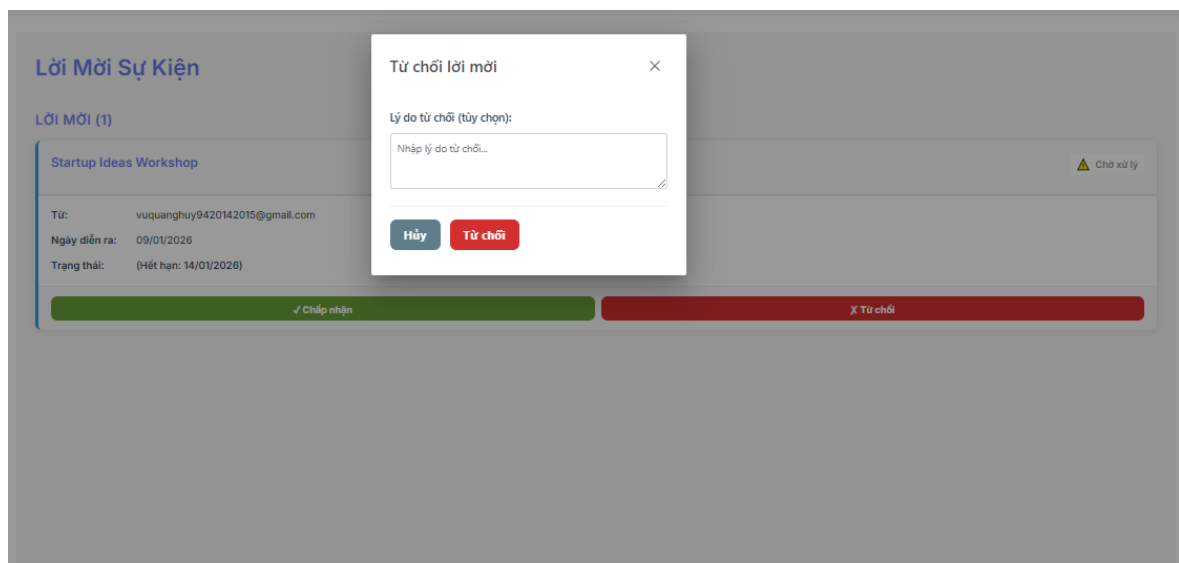
Từ: vuquanghuy9420142015@gmail.com

Ngày diễn ra: 09/01/2026

Trạng thái: (Hết hạn: 14/01/2026)

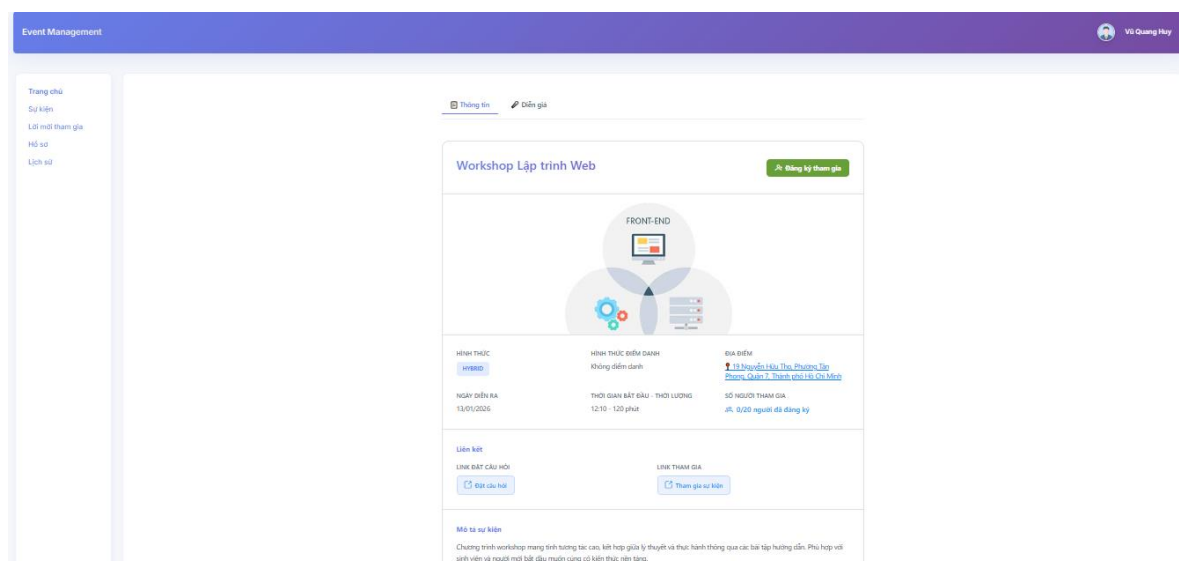
✓ Chấp nhận

X Từ chối



4.6 Page Register event

Students access the page, click on an event, and select an event to view its details. The event details and a registration button will then be displayed.



Students who register for an event will see information tabs supporting the event, such as their attendance history, resources, comments, and attendance times on the system (depending on whether the event has automatic attendance).

Students can cancel an event up to 2 days before it starts.

Thông tin

Điểm giá

Lịch sử điểm danh

Tài nguyên

Đánh luận

Workshop Lập trình Web

Đã đăng ký

Hủy đăng ký

FRONT-END



HÌNH THỨC

HYBRID

HÌNH THỨC ĐIỂM DANH

Không điểm danh

ĐỊA ĐIỂM

19 Nguyễn Hữu Thọ, Phường Tân Phong, Quận 7, Thành phố Hồ Chí Minh

NGÀY DIỄN RA

13/01/2026

THỜI GIAN BẮT ĐẦU - THỜI LƯỢNG

12:10 - 120 phút

SỐ NGƯỜI THAM GIA

1/20 người đã đăng ký

Liên kết

LINK ĐẶT CÂU HỎI

Đặt câu hỏi

LINK THAM GIA

Tham gia sự kiện

Mô tả sự kiện

Chương trình workshop mang tính tương tác cao, kết hợp giữa lý thuyết và thực hành thông qua các bài tập hướng dẫn. Phù hợp với sinh viên và người mới bắt đầu muốn củng cố kiến thức nền tảng.

In addition, when registering for an event, students also receive event notification emails such as registration confirmation emails, event update emails, event cancellation emails, etc.

Xin chào Vũ Quang Huy,

Cảm ơn bạn đã đăng ký tham gia sự kiện **Full-Stack Development Workshop**.

 **Thời gian điểm danh**

Check-in: 11:50 - 12:05

Check-out: 12:35 - 12:50

Vui lòng quét mã QR bên dưới khi đến tham dự sự kiện:

Mã QR điểm danh của bạn



 **Ngày:** 2026-01-22

Depending on whether the event uses QR code scanning or automatic attendance tracking, send the appropriate email.

In this case, the event supports QR code scanning and automatic attendance tracking, so users receive the time and QR code.

Xin chào Vũ Quang Huy,

Bạn đã hủy vé tham gia sự kiện **Full-Stack Development Workshop**.

 Ngày: 2026-01-22

 Giờ: 11:50

 Địa điểm: 6B Đường Chánh Hưng, Xã Bình Hưng, Huyện Bình Chánh, Thành phố Hồ Chí Minh

Nếu bạn muốn đăng ký lại, vui lòng truy cập trang sự kiện trước 2 ngày diễn ra.

Trân trọng,
Hệ thống Event Management

Mail send when the user unsubscribe the event

Xin chào Vũ Quang Huy,

Sự kiện bạn đã đăng ký **Full-Stack Development Workshop** vừa được cập nhật:

-  Ngày cũ: 2026-01-22 → Ngày mới: 2026-01-07
-  Giờ cũ: 11:50 → Giờ mới: 11:50
-  Địa điểm cũ: 6B Đường Chánh Hưng, Xã Bình Hưng, Huyện Bình Chánh, Thành phố Hồ Chí Minh → Địa điểm mới: 6B Đường Chánh Hưng, Xã Bình Hưng, Huyện Bình Chánh, Thành phố Hồ Chí Minh

Mail send when the organizer edit the event

Xin chào Vũ Quang Huy,

Chúng tôi rất tiếc phải thông báo rằng sự kiện **Vũ Quang Huy** đã bị **hủy bỏ**.

Mọi thông tin về và lịch trình liên quan đến sự kiện này hiện không còn hiệu lực.

Nếu có bất kỳ thắc mắc nào, vui lòng liên hệ ban tổ chức để được hỗ trợ.

Trân trọng,
Hệ thống Event Management

Mail send when the organizer cancel the event

4.7 Page Scan QR code

The event organizers/managers select the QR code scanning function to open the camera and take attendance.

Registered participants use the QR code they received from their registration email to check in.

Quét mã điểm danh

Sự kiện

Full-Stack Development Workshop

Quét mã QR

Mở camera

Danh sách điểm danh

STT	Email	Tên	MSSV	Check-in Time	Check-out Time	Trạng thái
Chưa có bản ghi điểm danh						
				<< < > >>	10	

✔️ Quét mã thành công

Mã Ticket: 7e2ca4bd-a8df-4cb4-ad24-4842cfff49abd

Tên người dùng: Vũ Quang Huy

Trạng thái: Check-in

Danh sách điểm danh

STT	Email	Tên	MSSV	Check-in Time	Check-out Time	Trạng thái
1	vuquanghuy811897@gmail.com	Vũ Quang Huy	S18H0199	15:18:11	-	✔️ Checked In

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After successful attendance check, the information of the person checking in and the attendance status will be displayed. Below that, the attendance information of all those checking in will be shown.

Điểm danh hôm nay

Điểm danh sự kiện: Full-Stack Development Workshop

Mở check-in: 12:50 - 14:05 | Mở Check-out: 16:35 - 15:50

Check-in vào lúc: 13:17

Đã check-in

Check-out

The attendance system incorporates a user location check function to prevent attendance fraud

Trang chủ

Sự kiện

Lời mời tham gia

Hồ sơ

Lịch sử

■ Điểm danh hôm nay

Điểm danh sự kiện: Full-Stack Developer

Mở check-in: 15:25 - 15:40 | Mở Check-out: 16:00

Đang lấy vị trí của bạn



Đang lấy vị trí của bạn...

1

Mở

Check-out

CHAPTER 5. CONCLUSION

5.1 Conclusion and Achievements

The system has completed the development of core functions, meeting the basic operational requirements in the event organization and management process, from event creation, invitation, ticket management, check-in/out, to surveys and reporting. The integration of attendance tracking methods provides users with flexibility and reduces the time spent on manual attendance taking.

5.2 Disadvantage

The system still has some shortcomings that need to be addressed. Firstly, it does not yet support multiple languages, making it difficult for international users. Secondly, the system only operates on a web platform and lacks a mobile application, limiting accessibility. This reduces the accuracy of the automatic attendance system because it requires knowing the user's location to prevent fraud.

5.3 Difficulties encountered

During the project implementation, the development team encountered several significant difficulties and limitations. Familiarizing themselves with and using TypeScript proved challenging, especially for new members, due to the need to understand concepts such as generics, interfaces, types, decorators, and TypeScript configuration. This resulted in significant time spent defining complex data types and handling type-related errors. Furthermore, the team did not effectively utilize TypeScript's advanced features to optimize and improve code maintainability. In addition, limitations in programming thinking and system design also affected code quality, resulting in difficulties in problem-solving, duplicate code, inefficient asynchronous handling, and potential errors during development.

CHAPTER 6. TÀI LIỆU THAM KHẢO

<https://spring.io/projects/spring-boot.com>

[Hoc Spring Boot Bắt Đầu Từ Đâu? - Viblo](#)

[Quick Start – React](#)

[Geolocation API - Web APIs | MDN](#)

[TrackAsia - MAP API Solutions](#)